# EDMeta: Example

## 1. Loading Data

We illustrate EDMeta using data from a published meta-analysis of long-term $PM_{2.5}$ exposure and all-cause mortality (Chen and Hoek, 2020), which includes $n = 25$ cohort study-specific effect estimates and standard errors, each standardized to a 10 $\mu g/m^3$ increase in $PM_{2.5}$, along with four exposure distribution summary metrics for each study: lower bound (`min_exposure`), upper bound (`max_exposure`), central value (`mid_exposure`), and standard deviation (`sd_exposure`).

```
load(url("https://raw.githubusercontent.com/warrenjl/EDMeta/main/EDMeta_Example/Chen_Hoek_2020.RData"))
```

## 2. Fitting the Model

We fit EDMeta using a truncated normal exposure distribution (`exposure_distribution = 1`) for each study, parameterized by the reported central value and standard deviation and truncated to the reported exposure range.

```
results <- EDMeta(mcmc_samples            = 110000,
                  beta_hat                = final$estimate,
                  se                      = final$se,
                  min_exposure            = final$min_exposure,
                  max_exposure            = final$max_exposure,
                  mid_exposure            = final$mid_exposure,
                  sd_exposure             = final$sd_exposure,
                  exposure_distribution   = 1,
                  metrop_sd_phi_eta_trans = 0.75)
```

## 3. Analyzing Output

```
# Burn-in and thinning
burnin   <- 10000
thin     <- 10
keep_set <- seq(burnin + 1, length(results$mu), by = thin)

# Posterior summaries for scalar parameters
params <- data.frame(
  Parameter = c("tau2", "sigma2_eta", "phi_eta"),
  Mean      = round(c(mean(results$tau2[keep_set]),
                      mean(results$sigma2_eta[keep_set]),
                      mean(results$phi_eta[keep_set])), 3),
  Lower95   = round(c(quantile(results$tau2[keep_set],     0.025),
                      quantile(results$sigma2_eta[keep_set], 0.025),
                      quantile(results$phi_eta[keep_set],  0.025)), 3),
  Upper95   = round(c(quantile(results$tau2[keep_set],     0.975),
                      quantile(results$sigma2_eta[keep_set], 0.975),
                      quantile(results$phi_eta[keep_set],  0.975)), 3)
)
print(params)
```

```
##    Parameter  Mean Lower95 Upper95
## 1       tau2 0.004   0.002   0.008
## 2 sigma2_eta 0.392   0.005   2.890
## 3    phi_eta 0.624   0.008   2.742
```

```r
# Metropolis acceptance rate for phi_eta
cat("phi_eta acceptance rate:",
    round(100 * results$acctot_phi_eta_trans / (110000 - 1)), "%\n")
```

```
## phi_eta acceptance rate: 23 %
```

```r
# Construct prediction grid and compute posterior exposure-response curve
n      <- nrow(final)
m      <- 100
m_star <- 100

a_star <- min(final$min_exposure)
b_star <- max(final$max_exposure)

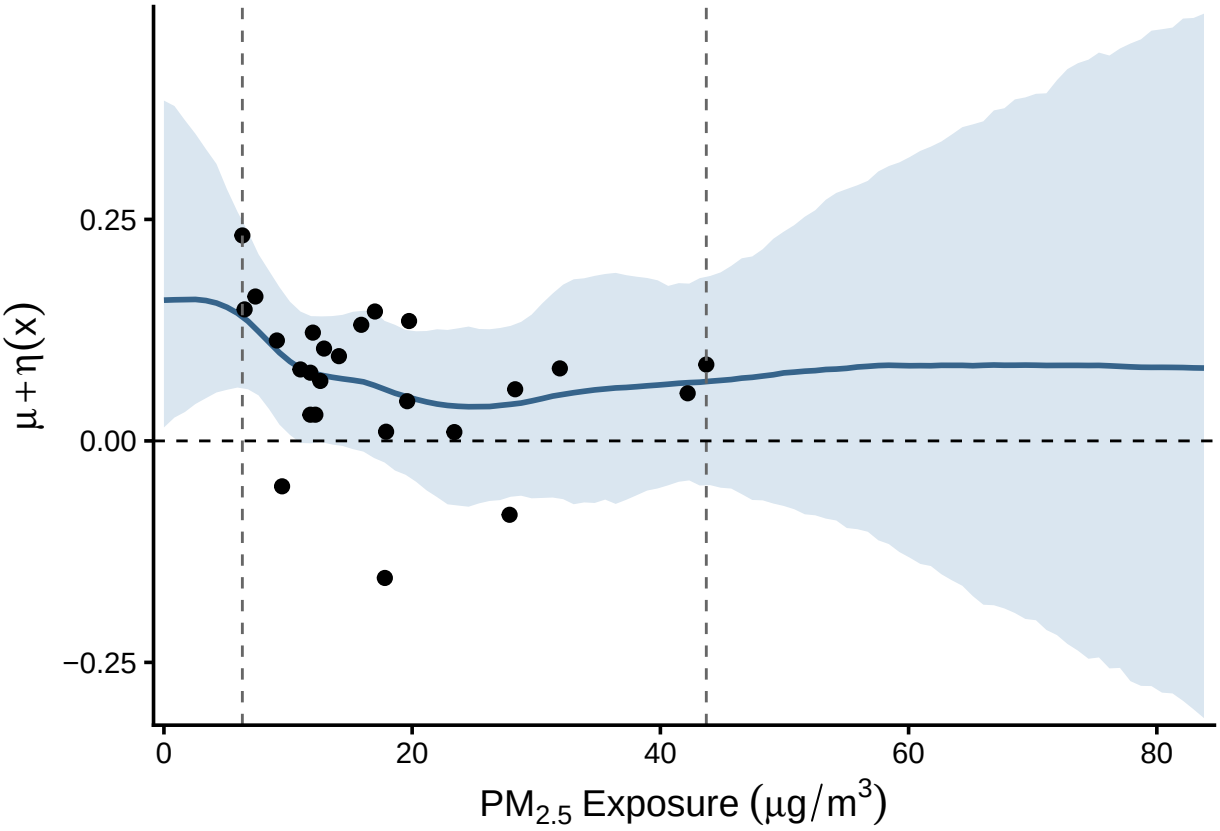
d_star <- seq(a_star, b_star, length.out = m_star)
d_pred <- seq(a_star, b_star, length.out = m)

abs_d_star_diffs  <- abs(outer(d_star, d_star, "-")) / b_star
abs_d_diffs_pred  <- abs(outer(d_pred, d_star, "-")) / b_star

curve <- matrix(NA, nrow = m, ncol = length(keep_set))
for (j in 1:length(keep_set)) {
  Sigma_inv <- chol2inv(chol(exp(-results$phi_eta[keep_set[j]] * abs_d_star_diffs)))
  x_pred    <- exp(-results$phi_eta[keep_set[j]] * abs_d_diffs_pred)
  curve[, j] <- results$mu[keep_set[j]] +
                x_pred %*% (Sigma_inv %*% results$eta_star[, keep_set[j]])
}

curve_summary <- data.frame(
  d_pred = d_pred,
  mean   = rowMeans(curve),
  lower  = rowQuantiles(curve, probs = 0.025),
  upper  = rowQuantiles(curve, probs = 0.975)
)

# Plot posterior exposure-response curve
ggplot(curve_summary, aes(x = d_pred, y = mean)) +
  geom_ribbon(aes(ymin = lower, ymax = upper),
              fill = "steelblue", alpha = 0.2) +
  geom_line(linewidth = 1, color = "steelblue4") +
  geom_point(data = final,
             aes(x = mid_exposure, y = estimate),
             shape = 19, size = 2, color = "black") +
  geom_vline(xintercept = min(final$mid_exposure),
             linetype = "dashed", color = "gray40", linewidth = 0.5) +
  geom_vline(xintercept = max(final$mid_exposure),
             linetype = "dashed", color = "gray40", linewidth = 0.5) +
  geom_hline(yintercept = 0,
             linetype = "dashed", color = "black", linewidth = 0.5) +
  labs(x = expression(PM[2.5]~Exposure~(mu*g/m^3)),
       y = expression(mu + eta(x))) +
  scale_x_continuous(expand = c(0.01, 0)) +
  scale_y_continuous(expand = c(0.01, 0)) +
  theme_classic(base_size = 14)
```


0.25
0.00
−0.25
0
20
40
60
80
μ+η(x)
PM2.5 Exposure (μg/m3)